

Companion LXI: Minkowski Functionals of Weak-Lensing Fields in the Relaxation-Driven Cyclic Cosmology

Michael Lehmann

`mi.lehmann@gmx.de`

April 2026

Abstract

Minkowski functionals provide a complete morphological and topological description of cosmological fields, capturing information beyond two- and three-point statistics. In weak-lensing analyses, they quantify the geometry of the convergence field, including area fractions, contour lengths, and Euler characteristics. In the standard cosmological model, these functionals are determined by the nonlinear matter distribution and the projection of large-scale structure. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the morphology, topology, and connectivity of lensing fields in a characteristic way. This Companion develops a continuous description of Minkowski functionals in RDCC, deriving how the relaxation scale influences the geometry of κ -fields, the topology of excursion sets, and the observational signatures in weak-lensing surveys. The resulting framework provides a distinctive probe of the RDCC gravitational field beyond correlation functions.

1 Introduction

Weak-lensing convergence maps contain rich information about the gravitational field. While power spectra and bispectra capture statistical correlations, they do not fully describe the geometry or topology of the field. Minkowski functionals provide a complete morphological description of excursion sets and are sensitive to non-Gaussianity, nonlinear structure formation, and the connectivity of the gravitational potential.

In the standard cosmological model, the Minkowski functionals of κ -fields reflect the nonlinear matter distribution and the projection of large-scale structure. In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the decay of small-scale modes and enhances the coherence of large-scale modes. This modifies the morphology, topology, and connectivity of κ -fields in a scale-dependent manner, producing a distinctive signature in Minkowski functional analyses.

2 Minkowski Functionals of the Convergence Field

For a 2D field such as the convergence $\kappa(\hat{\mathbf{n}})$, the Minkowski functionals of an excursion set $\kappa > \nu\sigma_\kappa$ are:

$$V_0(\nu) = \text{area fraction}, \tag{1}$$

$$V_1(\nu) = \text{contour length}, \tag{2}$$

$$V_2(\nu) = \text{Euler characteristic}. \tag{3}$$

These functionals encode:

- the abundance of high- χ regions,
- the geometry of contours,
- the topology of connected structures.

3 RDCC Modification of the Convergence Field

In RDCC, the convergence evolves as

$$\kappa_{\text{RDCC}}(\ell) = \kappa(\ell) \left[1 - \chi_{\kappa} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (4)$$

The variance becomes

$$\sigma_{\kappa, \text{RDCC}}^2 = \sigma_{\kappa}^2 \left[1 - \chi_{\sigma} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (5)$$

These modifications alter the morphology of excursion sets.

4 Minkowski Functionals in RDCC

The RDCC Minkowski functionals become

$$V_{0, \text{RDCC}}(\nu) = V_0(\nu) \left[1 - \chi_0 \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right], \quad (6)$$

$$V_{1, \text{RDCC}}(\nu) = V_1(\nu) \left[1 - \chi_1 \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right], \quad (7)$$

$$V_{2, \text{RDCC}}(\nu) = V_2(\nu) \left[1 - \chi_2 \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (8)$$

These modifications reflect the relaxation-driven changes in morphology and topology.

5 Topological Signatures of RDCC

The relaxation mechanism enhances the coherence of large-scale modes. Minkowski functionals therefore exhibit:

- larger connected high- χ regions,
- smoother contours and reduced small-scale curvature,
- a shift in the Euler characteristic zero-crossing,
- a characteristic turnover near ℓ_{rel} ,
- modified topology of voids and ridges.

These signatures provide a direct observational probe of the RDCC gravitational field in real-space morphology.

6 Conclusion

Minkowski functionals of weak-lensing fields in the Relaxation-Driven Cyclic Cosmology reflect the scale-dependent evolution of the gravitational potential and the nonlinear matter distribution. The relaxation mechanism modifies the geometry, topology, and connectivity of κ -fields in a scale-dependent manner, producing distinctive signatures in weak-lensing surveys. These effects provide a direct observational window into the relaxation scale and the temporal structure of the RDCC gravitational field. The present Companion establishes the theoretical foundation for future studies of topological cosmology, nonlinear structure, and the interplay between light and gravity in RDCC.

References

- [Mecke *et al.*(1994)Mecke, Buchert, and Wagner] K. R. Mecke, T. Buchert, and H. Wagner, *Astronomy and Astrophysics* **288**, 697 (1994).
- [Schmalzing and Buchert(1997)] J. Schmalzing and T. Buchert, *The Astrophysical Journal Letters* **482**, L1 (1997).
- [Matsubara(2003)] T. Matsubara, *The Astrophysical Journal* **584**, 1 (2003).
- [Hikage *et al.*(2008)Hikage *et al.*] C. Hikage *et al.*, *Monthly Notices of the Royal Astronomical Society* **389**, 1439 (2008).
- [Petri *et al.*(2013)Petri, Haiman, and Hui] A. Petri, Z. Haiman, and L. Hui, *Physical Review D* **88**, 123002 (2013).
- [Collaboration(2022)] D. Collaboration, *Physical Review D* **105**, 083519 (2022).
- [Collaboration(2021)] K. Collaboration, *Astronomy & Astrophysics* **646**, A140 (2021).
- [Collaboration(2023)] H. Collaboration, *Publications of the Astronomical Society of Japan* **75**, 1 (2023).
- [Park *et al.*(2005)Park, Kim, and Gott] C. Park, J. Kim, and J. R. Gott, *The Astrophysical Journal* **633**, 1 (2005).
- [Codis *et al.*(2012)Codis *et al.*] S. Codis *et al.*, *Monthly Notices of the Royal Astronomical Society* **427**, 3320 (2012).